The index/2 function is called whenever the URL http://localhost:4004/messages is loaded. It's called with a Plug.Conn struct that includes everything about the request, and the params provided for the request. If the conversation list has a conversation, then the request will be routed by the redirect/2 function to the detail page for that message. If the conversation list doesn't have any messages, then the request will redirect to a form to send a new message.

The code path = ~p(/messages/#{contact.id}) uses an important concept called verified routes.[15] The ~p sigil verifies that the provided route is present in your application and can be used. It would emit a warning if you were to replace the path with ~p(/nope).

Start the app with iex -S mix phx.server and navigate to http://localhost:4004/messages. Your browser will redirect to the detail page. We haven't defined it yet, so Phoenix shows the error "function PhoneAppWeb.MessageController.show/2 is undefined or private." If you get redirected to /messages/new, make sure that you run the seed functions from the previous section.

Next, let's define the show action.

Serve HTML via a Controller Action

The show action will be the main view for our user interface. It will display the selected conversation and will present a form to send a new SMS message.

To achieve this behavior, our action will need to load the contact, load the conversation with that contact, and assign a changeset to power the form. Add the following code to the MessageController module:

phone_app/lib/phone_app_web/controllers/message_controller.ex

```
def show(conn, params = %{"contact_id" => contact_id}) do
  contact = PhoneApp.Conversations.get_contact!(contact_id)
  conversation = PhoneApp.Conversations.load_conversation_with(contact)

  conn
  |> assign(:conversation, conversation)
  |> assign(:changeset, changeset(params))
  |> render("show.html")
end

defp changeset(params) do
  conversation_params = Map.get(params, "message", %{})
  PhoneApp.Conversations.new_message_changeset(conversation_params)
end
```

15. https://hexdocs.pm/phoenix/routing.html#verified-routes